

PROCEDURE

The project implements a system that selects the most functionally appropriate object for specific human tasks using the COCO task benchmarks. Unlike the standard tasks, our proposed project pipeline completely decouples spatial localization from semantic task-relevance, this eliminates non-deterministic post-processing and the heavy computation mathematics. This serves as a smooth structure for future FPGA acceleration.

PROBLEM WITH DATASET

The COCO dataset was made by human annotators, and they need to pick the correct tool from a limited, pre-set list of everyday objects for a specific task. So, these non-obvious annotation patterns directly influenced our design and sample image selection.

Task-13 (pound a carpet): The preferred objects are baseball bats and tennis rackets.

Task-12 (extinguish fire): Annotators preferred water bottles over fire hydrants, as they require specialist operation.

Task-7 (open bottle of beer): Dining tables and chairs are being marked as preferred objects, completely baffling a simple approach.

Task-6 (dig hole): Unusual objects like frisbee are preferred in a dataset with no objects like shovels, spades present.

So, to ensure our pipeline was tested against these real-world samples, we hand selected sample images for each such task.

PIPELINE ARCHITECTURE

STAGE 1: It employs YOLOv26n. Standard object detectors introduce heavy arithmetic bottlenecks that hurt latency and create routing challenges on FPGAs. YOLO26 removes Distribution Focal Loss (DFL), which eliminates the expensive SoftMax operation from the box decoding path entirely, simplifying inference and broadening hardware compatibility. It is also natively end-to-end NMS-free — predictions are generated directly through the forward pass with no separate post-processing suppression step, reducing latency and making the inference graph fully static and deterministic.

STAGE 2: Raw visual output is converted into a structured 180-dimensional feature space, making each transformation as lightweight as possible. The feature vector consists of a class one-hot (80 dims), task one-hot (14 dims), detection confidence (1 dim), bounding-box geometry (5 dims), and a scene presence vector (80 dims). The scene presence vector is computed once per image across all detections, giving the scorer global awareness of what objects are available in the scene — enabling context-sensitive selection such as preferring a cup only when no wine glass is detected.

STAGE 3: The first two layers use **Hardswish()** as it allows for smooth gradients keeping MLP free of exponential operations throughout. Sigmoid is applied by loss during training and training is performed on GPU, only inference runs on CPU.

Final Scoring: Multiplying by YOLO detection confidence ensures a highly-relevant but low-confidence detection is not over-ranked relative to a well-detected object.

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[Task 02] sit comfortably
→ couch (score: 0.788)
[Task 03] place flowers
→ vase (score: 0.925)
[Task 04] get potatoes out of fire
→ fork (score: 0.775)
[Task 05] water plant
→ bottle (score: 0.486)
[Task 06] get lemon out of tea
→ spoon (score: 0.925)
[Task 07] dig hole
→ frisbee (score: 0.873)
[Task 08] open bottle of beer
→ chair (score: 0.779)
[Task 09] open parcel
→ scissors (score: 0.752)
[Task 10] serve wine
→ wine glass (score: 0.944)
[Task 11] pour sugar
→ spoon (score: 0.705)
[Task 12] smear butter
→ knife (score: 0.896)
[Task 13] MISSING: sample_images/13.jpg
[Task 14] pound carpet
→ tennis racket (score: 0.806)
=====
Done. 13/14 tasks. Outputs in: outputs/
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C:\Users\PUKHAR\Desktop\DVCon\DVCon 2026 Project\stage_2A>
```

